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Discriminates Delay and Acceleration Framing in Intertemporal Choice

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Abstract: People often discount future rewards, embracing smaller rewards that are delivered sooner rather than waiting for larger rewards delivered later. Previous behavioral research has demonstrated that people are more patient when options are presented as decisions to accelerate rather than delay consumption. This behavioral effect is well-established in the literature, but the underlying neural mechanisms have not been identified. We examined the neural correlates of delay and acceleration framing in intertemporal choice. We find greater activation in the hippocampus, amygdala, and anterior insula when options were framed as decisions to delay rather than accelerate consumption. These findings are consistent with theoretical accounts that posit that preferences are constructed. Specifically, the heightened activation observed in medial temporal regions may reflect more vivid representations of sooner outcomes in delay versus acceleration framing. These results provide insight into contextual effects in intertemporal choice specifically and preference construction more broadly.

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Research Data Related to this Submission

Title: Framing the Future First: Medial Temporal Lobe Activation
Discriminates Delay and Acceleration Framing in Intertemporal Choice
Repository: Neurovault

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Please find attached our submission, “Framing the Future First: Medial Temporal Lobe Activation Discriminates Delay and Acceleration Framing in Intertemporal Choice,” for consideration for publication in *NeuroImage*.

Decisions about options that unfold over time are central to numerous aspects of daily life, from healthy eating to saving for retirement. Unfortunately, one of the hallmarks of these intertemporal choices is that people often discount future rewards, selecting lower valued options that are available sooner over higher value options available in the future. However, prior research has demonstrated that how intertemporal options are framed influences people’s patience. When the default option is the sooner, lower valued option (and the decision is therefore whether to delay consumption to the later option), people are less patient than when the later, higher valued option is the default (and the decision is therefore whether to accelerate consumption to the sooner option). This important effect of context on intertemporal choice is well-established in the behavioral literature, but its neural underpinnings remain uncharacterized. Clarifying the neural correlates of this framing effect would both inform understanding of the psychological processes underlying this effect and provide insight into how to promote patience in intertemporal choice.

The present research combines functional neuroimaging with an adapted intertemporal choice task. Participants underwent functional magnetic resonance imaging while making intertemporal choices that were framed either as decisions to accelerate or delay consumption. We observe greater activation in hippocampus, amygdala, and anterior insula when decisions are framed as decisions to delay consumption compared to when they are framed as decisions to accelerate consumption. This pattern of neural activation is consistent with theories of preference construction such as Query Theory, which posit that evidence favoring the default option should be retrieved first and may inhibit subsequent retrieval of information favoring the alternative option. The default is the smaller, sooner option in delay framing and the larger, later option in acceleration framing. Given that sooner events are represented more concretely and with more associative detail than later outcomes, it is therefore expected that more activation should be observed in regions involved in representing available outcomes in the delay rather than the acceleration frame, consistent with our results.

We suggest two reviewers for this work, each with relevant expertise. Joe Kable (kable@psych.upenn.edu) of University of Pennsylvania has relevant experience in researching intertemporal choice and decision neuroscience. John Monterosso (johnrmon@usc.edu) of University of Southern California has relevant experience in researching self-control and decision neuroscience.

We anticipate these findings will be of broad interest to researchers in cognitive neuroscience. To our knowledge, this is the first paper to examine the neural underpinnings of delay versus acceleration framing in intertemporal choice. In doing so, we advance understanding of the neurocognitive processes that support this behavioral effect and help promote patience. Additionally, this paper extends prior understanding of how preferences are constructed during intertemporal choice, with broad implications for theories of decision making. Importantly, these findings offer insights into how neurocognitive representations of options are activated during deliberation to generate preferences. Such insights both inform basic understanding of the

neurocognitive architecture underlying decision making and suggest potential means to help encourage more patient choice.

Sincerely,

Crystal Reeck, Bernd Figner, Elke Weber, Jason Steffener, Amy Krosch, Tor Wager, and Eric Johnson

Framing the Future First: Medial Temporal Lobe Activation Discriminates Delay and
Acceleration Framing in Intertemporal Choice

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Abstract

People often discount future rewards, embracing smaller rewards that are delivered sooner rather than waiting for larger rewards delivered later. Previous behavioral research has demonstrated that people are more patient when options are presented as decisions to accelerate rather than delay consumption. This behavioral effect is well-established in the literature, but the underlying neural mechanisms have not been identified. We examined the neural correlates of delay and acceleration framing in intertemporal choice. We find greater activation in the hippocampus, amygdala, and anterior insula when options were framed as decisions to delay rather than accelerate consumption. These findings are consistent with theoretical accounts that posit that preferences are constructed. Specifically, the heightened activation observed in medial temporal regions may reflect more vivid representations of sooner outcomes in delay versus acceleration framing. These results provide insight into contextual effects in intertemporal choice specifically and preference construction more broadly.

Keywords: Intertemporal Choice, Decision Neuroscience, Hippocampus, Framing Effects, Preference Construction.

1. Introduction

Intertemporal choice underpins important decisions humans face daily, from deciding whether to eat indulgent foods to saving for retirement. The tradeoff between lower-valued sooner options and more valuable future options determines many life outcomes (Golsteyn, Gronqvist, & Lindahl, 2014), and much research has focused on how to increase willingness to postpone immediate rewards. Most prior neuroscience research on intertemporal choice has focused on identifying neural correlates of parameters from computational discounting models (Kable & Glimcher, 2007; McClure, Laibson, Loewenstein, & Cohen, 2004).

Such discounting models, however, cannot account for a critical contextual effect in intertemporal choice—the well-documented asymmetry observed in discounting when options are framed as decisions to delay or accelerate consumption (Loewenstein, 1988; Malkoc & Zauberman, 2006; Reeck, Wall, & Johnson, 2017; Weber et al., 2007). When options are framed as decisions to delay consumption (with the sooner option as the default), people often make more impatient choices than when options are framed as decisions to accelerate consumption (with the later option as the default).

This difference likely emerges due to the constructed nature of preferences (Lichtenstein & Slovic, 2006). One account, Query Theory, suggests that people first retrieve reasons favoring the default option and then consider reasons for pursuing the alternative (Weber et al., 2007). Because of output interference, information favoring the first option considered comes to mind more easily and diminishes activation of arguments favoring alternative choices. This creates an asymmetry such that greater pro-default information is recalled. In delay, the default is the smaller, sooner option; in acceleration, it is the larger, later option. Because sooner events are represented more concretely (Malkoc & Zauberman, 2006) and involve retrieval of more vivid

detail than more distant future events (Schacter, Addis, & Buckner, 2007; Tamir & Mitchell, 2011), delay framing (i.e., smaller sooner default) should result in greater activation in regions involved in intertemporal preference construction than acceleration framing (i.e., larger later default). This differential activation occurs because the default option is typically considered first and, in the delay frame, this default is the smaller, sooner option which will be represented with more associative detail than the larger, later option is represented with in the acceleration frame. This asymmetry suggests that acceleration decisions may promote patience by diminishing representation of the smaller, sooner option by framing the future first.

While prior work has identified the neural regions that are involved in representing value during intertemporal choice (Kable & Glimcher, 2007; McClure et al., 2004), the neural correlates of delay and acceleration framing remain uncharacterized. Elucidating the neural correlates of this critical contextual effect on intertemporal choice would clarify the neurocognitive architecture that is involved in the construction of preferences. The present research sought to delineate neural activation in delay and acceleration framing in order to inform understanding of preference construction in intertemporal choice.

2. Methods

2.1 Participants

Twenty participants (12 women, mean age = 26.0 years old) participated in the experiment in exchange for payment. The sample size is commensurate with other neuroimaging studies of intertemporal choice (Kable & Glimcher, 2007; McClure et al., 2004). Participants were screened for contraindications for participating in MR scanning (e.g., implanted metal, claustrophobia) and were all right-handed. Imaging data from one participant were excluded

from all analyses due to excessive motion. All participants provided informed consent and all procedures were approved by the Columbia University Institutional Review Board.

2.2 Materials and Procedure

After providing informed consent, participants completed an intertemporal choice task. On each trial of this task (Figure 1), participants chose between a smaller amount of money that would be delivered sooner (SS, e.g., \$58.90 today) and a larger amount of money that would be delivered later (LL, e.g., \$72.90 in 2 weeks). A set of 72 different option pairs was constructed using a process similar to previous research involving intertemporal choice (McClure et al., 2004; Reeck et al., 2017). SS reward amounts were drawn from a distribution with a mean of \$45 and a standard deviation of \$20, constrained so that all amounts fell between \$15 and \$85. The set of choice pairs was then constructed by varying (i) the SS delivery time (today or in 2 weeks), (ii) the delay between the SS and LL delivery times (2 or 4 weeks delay), and (iii) the relative difference in the dollar amounts between the two options (1, 3, 5, 10, 15, 20, 25, 35, or 50%). Two option pairs were generated for each of the 36 combinations of these features, for a total of 72 unique choice trials.

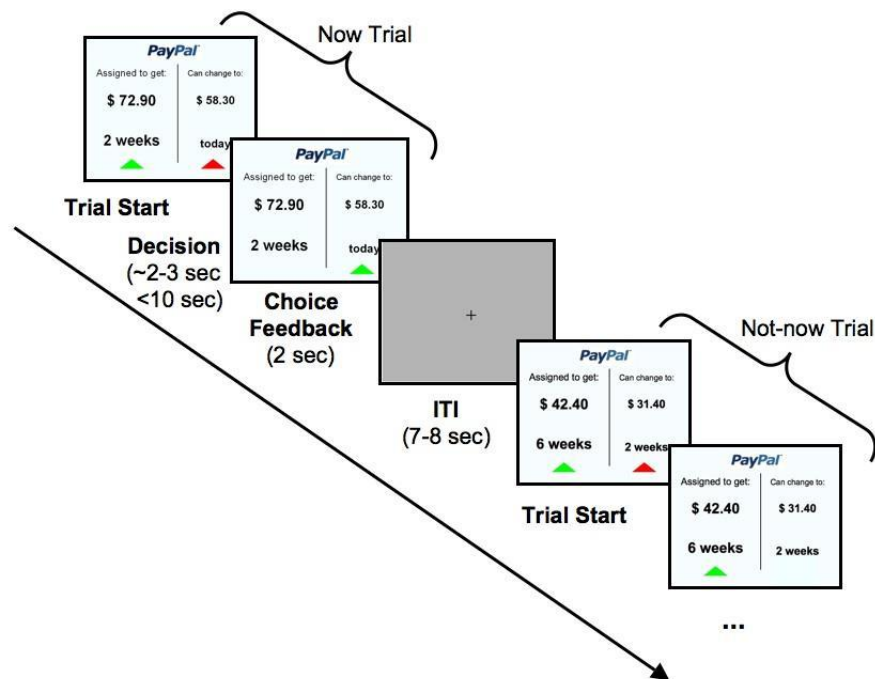


Figure 1

Two example trials from the intertemporal choice task. Each trial began with the presentation of two options, one of which was a smaller amount of money that was delivered sooner and the other of which was a larger amount of money delivered later. A triangle appeared beneath each option. Participants made their selection in a self-paced fashion, with a maximum of 10 seconds to deliberate on each trial. Once they made their selection, a green triangle was presented underneath their chosen option, and this display remained on the screen for two seconds. After a 7-8 second jittered inter-trial interval, the next trial began. Acceleration and Delay framed trials were presented in separate runs, and the gift certificate participants decided about also varied between runs.

The key manipulation in the present experiment was whether the choice was framed as a decision to delay or accelerate consumption by making either the SS or the LL option the default (Loewenstein, 1988; Malkoc & Zauberman, 2006; Reeck et al., 2017; Weber et al., 2007). When the SS reward was presented as the default option, switching to the LL reward would delay consumption. Conversely, when the LL reward was presented as the default option, switching to the SS reward would accelerate consumption. The default option was always presented on the left side of the display, printed in larger font, with a green triangle beneath it, and with text indicating that the participant was assigned to get this option. The alternate option was always presented on the right side of the display in smaller font with a red triangle beneath it and text indicating that the participant could change to this option (see **Figure 1**). The default was always presented on the left so that natural reading order would reinforce the left option (which would likely be read first) as the default. Note that there was no actual switching cost from the default to the alternative option since participants had to indicate either choice via button press. Participants' decisions were self-paced, with a maximum deliberation time of 10 seconds. Once participants made a selection, a green triangle appeared beneath the selected option for 2 seconds, followed by a 7-8 second jittered intertrial interval. Thus, the initial presentation of the green triangle beneath the left option also reinforced that this option was the default.

Delay versus acceleration framing was manipulated between functional scanning runs. Two runs featured delay framing and two featured acceleration framing, with the order counterbalanced across participants (delay-accelerate-accelerate-delay or accelerate-delay-delay-accelerate). To help participants differentiate between the two frames, the rewards for the two choice options were provided as gift certificates from different providers (Amazon and PayPal) counterbalanced across the two framing conditions. The same set of 72 option pairs was

presented in both the delay and accelerate conditions, and the trials presented in each run consisted of the same number of each of the 36 trial types. Participants initially practiced this task briefly outside the scanner, completing a few delay and acceleration trials. The duration of each functional run was fixed, such that participants completed on average 128 choice trials (range: 117 to 137 trials) while undergoing functional magnetic resonance imaging. To ensure that the task was incentive-compatible, participants were informed that one of these choices would be randomly selected and they will receive the gift-certificate reward they chose on that trial, delivered after the experiment via email (but redeemable only after the trial-specified delay).

2.3 Behavioral Analyses

The main model analyzed choice—smaller, sooner (SS) versus larger, later (LL)—as a function of framing (acceleration or delay), the immediacy of the SS option (available Now versus Not Now), the interaction between framing and SS immediacy, the time difference between the SS and LL (2 or 4 weeks), the relative difference in reward magnitudes between the SS and LL, and the reward magnitude of the SS. In addition to these fixed effects, we included a random intercept varying over participants, random slopes varying over participants for each of the fixed effects, as well as all possible pairwise covariances between the random effects. Thus, this model constitutes a "maximal model" with respect to the random effects, as recommended by Barr, Levy, Shchepers, and Tily (2013) to safeguard against inflated Type 1 errors. All categorical predictors (framing and SS immediacy) were sum-to-zero coded. All the other, continuous predictors were standardized. The models were implemented using the package *brms* (Bürkner, 2017) in R (Team, 2019), which provides an interface to Stan (Carpenter et al., 2016). All models were run with 6 chains with each 2000 iterations (1000 of which were warm-up). We

used brms's default priors. All models were checked for convergence via the Rhat values and visual inspection of the trace plots. We deemed a regression coefficient statistically significant if its 95% posterior credible interval (CI) did not include 0 (for marginally significant effects, we inspected the 90% CI).

To better characterize the interaction between frame and immediacy we report in the main text, we ran four follow-up models, analyzing choice separately for (i) now trials, (ii) not-now trials, (iii) delay-framed trials, and (iv) acceleration-framed trials. These models were identical to our main choice model, except that they were run always on the respective subset of the data and always excluded the relevant respective predictor (i.e., the separate models for the now trials and the not-now trials did not include the immediacy predictor; similarly, the separate models for the delay-framed and the acceleration-framed trials did not include the frame predictor).

Response times were analyzed using a similar model as that employed for choice data, using the same fixed and random effects as our main choice model. The dependent variable was the trial-level response times (RT). Since they showed a non-normal distribution, we used a generalized linear mixed-effects model with a shifted lognormal distribution, which provided a good fit to our data.

2.4 Image Acquisition and Analysis

Functional and anatomical images were acquired with a 1.5 T twin-speed GE MRI scanner using an echo-planar pulse sequence and an eight-channel head coil (TR = 2.0 s, TE = 34 ms, flip angle = 90°). Twenty-nine contiguous axial slices covering the full brain were acquired along the AC-PC plane, with a 64 x 64 matrix and 22.4 cm FOV (slice thickness = 3.5 mm, slice

separation = 0 mm). High-resolution structural images were acquired using a 3D SPGR sequence (124 slices, matrix size = 256 x 256, FOV = 22.0 cm).

Functional MRI data were analyzed using Statistical Parametric Mapping software (SPM8, Friston et al., 1995). The first five acquisitions were discarded to allow for signal stabilization. Functional images for each participant were corrected for slice acquisition timing, realigned to correct for motion, and co-registered with the participant's high-resolution structural scan. Each participant's bias-corrected high-resolution structural image was normalized to the Montreal Neurological Institute ICBM 152 template using unified segmentation (Ashburner & Friston, 2005). The resulting transformation parameters were then applied to the functional images, which were spatially smoothed using a Gaussian kernel of full-width half-maximum 8.0 mm³. Functional images had a final spatial resolution of 3.5 mm x 3.5 mm x 3.5 mm.

A general linear model was constructed for each participant with a total of four regressors modeling task effects. Separate regressors were constructed for the trial onsets of each condition as a function of the framing (delay or acceleration) and the delivery timing of the smaller, sooner reward (today or 2 weeks). For each task effect regressor, two additional regressors were included to model parametric effects of the trial options' subjective value and the participant's response time. To generate estimates of subjective value for each trial, we estimated each participant's individual discount rate. As the overwhelming majority of the literature reports better fits for hyperbolic than exponential discounting models (Green & Myerson, 2004), we used Mazur's (1987) standard one-parameter model of hyperbolic discounting: Subjective Value = Objective Amount / (1 + $k \times \text{Delay}$), where *Delay* indicates the time of delivery (in years) and k is a constant that is specific to each participant and indicates the steepness of the participant's discounting (with larger values indicating steeper discounting and 0 indicating no discounting at

all). For each participant, the hyperbolic discount rate k best explaining their choice pattern was estimated by fitting a logistic curve to their choices (plotting choice of SS versus LL as a function of the indifference-implied discount rate of each choice pair). The participant's estimated discount rate was thus the discount rate at which the participant would be predicted to choose the SS and LL with equal probability. The discount rate was used in the fMRI analysis to compute a regressor reflecting the subjective discounted value of the choice options presented in a trial. The results presented here used for this regressor the sum of the subjective discounted values of the SS and the LL. In supplemental analyses not presented in this paper, we used instead (i) the difference in subjective discounted values between the SS and LL and (ii) two separate values for k (one for Delay and one for Acceleration) to compute the sum and the difference regressors. For each of these variations, the results remained qualitatively the same. Time points were convolved with the canonical hemodynamic response, and linear contrasts between conditions of interest were estimated for each participant and subsequently employed in second-level random effects analyses to obtain mean t -images. SnPM (Winkler, Ridgway, Webster, Smith, & Nichols, 2014) was employed to compute a combined cluster-extent and voxel-height threshold to correct for multiple comparisons. A voxel-height threshold of $p < .01$ was employed to generate corrected thresholds with a family-wise error rate of $p < .05$ in the analyses reported below. Data are available at <https://identifiers.org/neurovault.image:305023>

2.5 Eye-tracking Analysis

For fifteen participants, eye-tracking data were collected while they performed the temporal discounting task in the scanner. Individual differences in search strategies during decision making have been shown to influence patience (Reeck et al., 2017), but it remains unclear whether differences in search strategies also alter neural processes during decision

making. For each trial, we calculated a Payne Index to summarize search behavior (Payne, 1976). Payne Index is the relative difference between alternative-based and attribute-based transitions $[(\text{Alternative} - \text{Attribute})/(\text{Alternative} + \text{Attribute})]$. Those who engage in more comparative searching (*e.g.*, looking at the monetary amounts of each option before then looking at the delivery time of each option) have negative values while those who engage in integrative searching (*e.g.*, looking at the monetary amount and delivery time for one option before moving on to the next option) have positive values. An average Payne Index was calculated for each participant and entered as a second-level covariate in analyses to identify regions whose activation tracked individual differences in search strategies.

3 Results

3.1 Behavioral Results

We used a Bayesian mixed-effects model to analyze choice as a function of framing (acceleration or delay), the immediacy of the smaller, sooner (SS) option (available Now versus Not-now), the interaction between framing and SS immediacy, the time difference between the SS and the larger later (LL) option (2 or 4 weeks), the relative difference in reward magnitudes between the SS and LL, and the reward magnitude of the SS. The main effects of frame and immediacy on choice were non-significant ($B = 0.05$, $s.e. = 0.14$, 95% CI $[-0.21, 0.33]$; $B = 0.25$, $s.e. = 0.22$, 95% CI $[-0.19, 0.70]$, respectively). There was, however, a significant interaction between framing and immediacy, $B = -0.35$, $s.e. = 0.12$, 95% CI $[-0.60, -0.12]$, see **Figure 2**. To better characterize the interaction between frame and immediacy, we ran separate follow-up models for (a) Now and Not-now trials and (b) for delay-framed and acceleration-framed decisions. In Now trials, we found participants made significantly more patient choices in acceleration compared to delay framing, $B = 0.35$, $s.e. = 0.15$, 95% CI $[0.06, 0.67]$, consistent

with prior literature (Loewenstein, 1988; Malkoc & Zauberman, 2006; Reeck et al., 2017; Weber et al., 2007). There was no significant difference in Not-now trials, $B = -0.33$, $s.e. = 0.25$, 95% CI $[-0.84, 0.13]$. Similarly, in delay-framed decisions, we observed a significant immediacy effect, $B = 0.59$, $s.e. = 0.28$, 95% CI $[0.05, 1.16]$, with more impatient choice when an immediate reward was available compared to when all rewards were in the future. In acceleration-framed decisions, the immediacy effect was not significant, $B = -0.10$, $s.e. = 0.25$, 95% CI $[-0.58, 0.40]$. In addition, we found significant effects in the expected directions for relative reward differences, $B = -0.35$, $s.e. = 0.12$, 95% CI $[-0.60, -0.12]$, SS amount, $B = -0.35$, $s.e. = 0.12$, 95% CI $[-0.60, -0.12]$, and time difference, although this latter effect was only marginally significant, $B = -0.35$, $s.e. = 0.12$, 90% CI $[-0.51, -0.04]$.

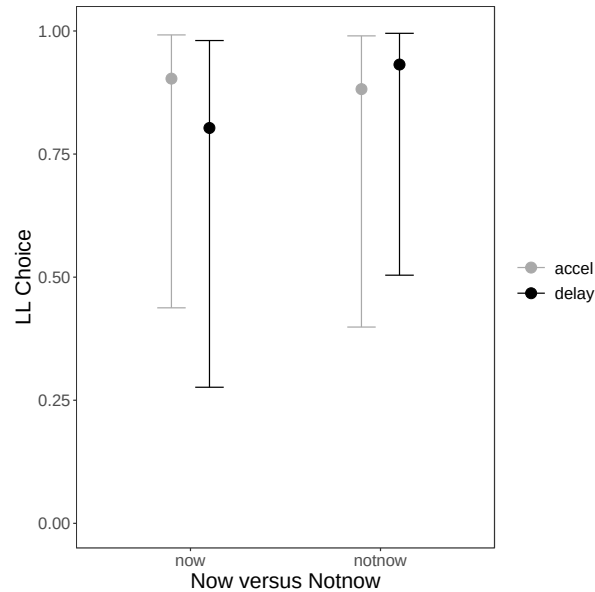


Figure 2

Proportion of larger, later (LL) choices as a function of frame (acceleration vs. delay) and immediacy (Now versus Not-now trials). We observed a significant framing x immediacy interaction such that in the Now trials, participants were significantly more patient in acceleration than delay decisions, while there was no significant framing effect for Not-now trials. The figure represents a marginal effects plot with error bars indicating 95% CIs.

To investigate whether framing and/or immediacy affected response time (RT), we ran a model similar to the choice model reported above, but with RT as the dependent variable. The only significant effect was that participants were slightly faster in Now than Not-now trials, $B = 0.03$, $s.e. = 0.01$, 95% CI [0.003, 0.06], possibly due to somewhat higher processing demands when both time points were in the future. All other effects were non-significant (frame: $B = 0.02$, $s.e. = 0.02$, 95% CI [-0.02, 0.06]; relative amount difference: $B = -0.04$, $s.e. = 0.03$, 95% CI [-0.11, 0.03]; time difference: $B = -0.007$, $s.e. = 0.01$, 95% CI [-0.03, 0.01]; SS amount: $B = -0.03$,

$s.e. = 0.02$, 95% CI [-0.07, 0.02]; immediacy x frame: $B = 0.004$, $s.e. = 0.01$, 95% CI [-0.02, 0.02]).

3.2 Neural Correlates of Framing

Our central goal was to characterize the neural correlates of framing options in intertemporal choice, identifying neural regions exhibiting differential activation for delay versus acceleration. Whole-brain analyses revealed a network of regions that exhibited greater activation under delay compared to acceleration framing (**Figure 3**), including hippocampus, amygdala, and insula (**Table 1**). No regions were identified that exhibited greater activation for acceleration compared to delay. The greater observed activation in the medial temporal lobe in delay is consistent with specific constructed preferences accounts of asymmetric discounting. For example, Query Theory posits that the status quo option is considered first by participants, namely the smaller, sooner option in delay framing and the larger, later option in acceleration framing. Because sooner events are processed more concretely (Malkoc & Zauberan, 2006) and imagined with more vivid detail (Schacter et al., 2007; Tamir & Mitchell, 2011), greater activation should be observed in regions involved in imagining the options in the delay condition, where the sooner option gets considered first. Because larger, later options are more temporally distant than smaller, sooner options, there is less associative detail that comes to mind when larger, later options are considered first (in acceleration) compared to when smaller, sooner options are considered first (in delay). In both acceleration and delay, the alternative option is considered second, and thus comes to mind with less associative detail due to interference from the considered default option. Thus, richer associative detail is likely represented in the delay condition (due to primary consideration of the smaller, sooner option) compared to the acceleration condition. Since medial temporal lobe structures are centrally involved in imagining

future events and recalling detail (Peters & Buchel, 2011; Schacter et al., 2007; Tamir & Mitchell, 2011), their greater activation in the delay condition may reflect the greater vividness of smaller, sooner options when they are considered first compared to the acceleration condition where the more pallid larger, later options are considered first.

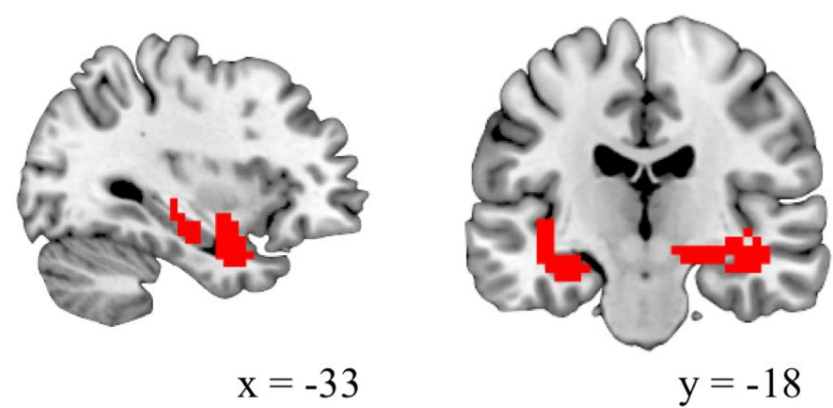


Figure 3

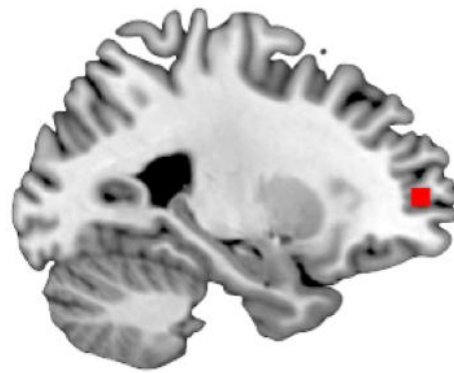
Neural correlates of option framing. Regions exhibiting greater activation for delay compared to acceleration framing are displayed. Whole-brain corrected, $p < .05$. MNI Coordinates for activation peak listed in **Error! Reference source not found..**

Table 1. Regions exhibiting greater activation for delay compared to acceleration framing.

Regions	MNI Coordinates (x, y, z)	Z	Cluster Size (voxels)
Hippocampus, Amygdala, Parahippocampal Gyrus, Fusiform Gyrus, Temporal Cortex, Insula, Occipital Cortex, Cerebellum	-34, 4, -23	3.91	1,088

3.3 Neural Regions Tracking Payne Index

Recent work has revealed that individual differences in search processes during intertemporal choice moderate the effect of option framing on intertemporal choice (Reeck et al., 2017). We sought to examine whether individual differences in search processes similarly moderated neural effects of option framing. Analyses examined which regions exhibited differences in acceleration versus delay framing that tracked Payne Index, a measure of search. No regions survived correction for multiple comparison corrections in whole-brain analyses. However, given recent interest in process models of intertemporal choice, we conducted an exploratory analysis to guide future researchers, using a threshold of $p < .001$ uncorrected, $k = 10$. A negative relationship between Payne Index and the magnitude of the neural framing effect was observed in the anterior prefrontal cortex (**Figure 4, Table 2**). Anterior prefrontal cortex has been previously implicated in intertemporal choice (Shamosh et al., 2008) and may be tied to choice difficulty and mentally representing different options (Jimura, Chushak, Westbrook, & Braver, 2018).



$x = -26$

Figure 4

Regions exhibiting a negative correlation between individual differences in Payne Index and the magnitude of the neural framing effect (Delay > Acceleration Framing). Uncorrected, $p < .001$, $k = 10$. MNI Coordinates for activation peaks listed in Table 2.

Table 2. Regions whose neural framing effect (Delay > Acceleration Framing) negatively correlates with individual differences in Payne Index.

Regions	MNI Coordinates (x, y, z)	Z	Cluster Size (voxels)
Anterior Prefrontal Cortex	-27, 52, 5	3.57	10

4. Discussion

Our goal was to delineate differences in neural activation between delay and acceleration frames in intertemporal choice. When options were framed as delays, we observe greater neural activation in the hippocampus, amygdala, and anterior insula—regions previously implicated in intertemporal choice. Anterior insula may play a role in projecting the self across time (Craig,

2009), an important process in delay discounting (Clewett et al., 2014). Animal models reveal that lesions to both the hippocampus (Cheung & Cardinal, 2005) and the amygdala (Winstanley, Theobald, Cardinal, & Robbins, 2004) result in more impatient decisions. Both regions have been implicated in episodic processing supporting intertemporal choice (Peters & Buchel, 2011). Medial temporal lobe structures are argued to play an essential role in supporting intertemporal choice by representing different available outcomes (Peters & Buchel, 2011), consistent with our results.

While we observed that people were more patient in the acceleration than the delay frame on trials featuring an immediate outcome, we did not observe a main effect of delay versus acceleration framing on choice. This effect, however, is well established behaviorally in the literature (Loewenstein, 1988; Malkoc & Zauberman, 2006; Reeck et al., 2017; Weber et al., 2007), but mostly in designs that manipulated the framing between participants and with many fewer trials. Recent work has revealed that individual differences in search processes during intertemporal choice moderate the effect of option framing on intertemporal choice (Reeck et al., 2017). It could be the case that these individual differences in the present sample resulted in an overall null effect of framing on behavior, but with a different composition of participants the traditional effect would emerge. Exploratory analyses of relationships between search processes during the task and activation in delay versus acceleration frames revealed activation in anterior prefrontal cortex. Anterior prefrontal cortex has been previously implicated in intertemporal choice (Shamosh et al., 2008) and may be particularly involved in representing options during deliberation (Jimura et al., 2018).

Although asymmetric discounting is well-established in the behavioral literature, the present research is the first examination of the neural basis for this observed difference in

intertemporal decisions. By characterizing the neural correlates of option framing in intertemporal choice, the present work advances general understanding of the neurocognitive processes that support this behavioral effect and promote patience. While neural data alone are insufficient to isolate the mechanisms underlying these phenomena, they provide valuable insight into potential processes contributing to such effects. Indeed, the present findings are consistent with a Query Theory account that emphasizes differences in retrieval in response to delay versus acceleration frames (Weber et al, 2007). Specifically, people recall more details favoring the default option, which is the smaller, sooner option in delay framing and the larger, later option in acceleration framing. Because sooner events are represented with richer associative detail (Malkoc & Zauberman, 2006; Schacter et al., 2007; Tamir & Mitchell, 2011), this asymmetry should result in greater activation in the delay frame than the acceleration frame, consistent with our findings.

Our findings also dovetail with a growing literature implicating the hippocampus in preference construction more broadly during decision making. Rodent experiments have demonstrated that the hippocampal activity during decisions predicts the selected environment (Johnson & Redish, 2007), consistent with the notion that hippocampus represents predictions of outcomes to support decision making (Johnson, van der Meer, & Redish, 2007). Hippocampal activation has also been linked to preference construction when people imagine new experiences that are based on combining familiar components (Barron, Dolan, & Behrens, 2013). Patients with hippocampal lesions have also been found to exhibit more inconsistencies in their choices from among a set of options (Enkavi et al., 2017), consistent with the theory that hippocampus helps represent the value of options during preference construction (Gluth, Sommer, Rieskamp, & Buchel, 2015). These findings implicate hippocampus in representing options' value during

decision making broadly, but medial temporal lobe has also been implicated in representing outcomes during intertemporal choice. The hippocampus is involved in imaging future episodic events during decision making (Benoit, Gilbert, & Burgess, 2011; Lebreton et al., 2013; Peters & Buchel, 2010), and hippocampal lesion patients fail to show benefits from episodic imagining of future outcomes during intertemporal choice (Palombo, Keane, & Verfaellie, 2015). The present results extend this prior work by showing that activation in medial temporal lobe structures also discriminates between delay and acceleration framing in ways predicted by theories of preference construction (Weber et al., 2007). Taken together, these findings suggest that the medial temporal lobe supports intertemporal choice through the representation of outcomes while people are deliberating between options and constructing their preferences.

Overall, the present findings implicate the medial temporal lobe and the anterior insula in constructing preferences during intertemporal choice. Differences observed between delay and acceleration framing of options indicate these neural regions may be especially important for representing different outcomes during decision making, consistent with previous theoretical accounts (Peters & Buchel, 2011; Weber et al., 2007). This has potential implications for possible interventions. Framing the future first appears to encourage a different imagining of options while preferences are constructed. This and other interventions based in value construction may thus provide a flexible means to encourage patience.

Author Contributions

B.F., E.U.W., and E.J.J. designed the experiment. B.F. and A.R.K. collected the data. C.R., J.S., and B.F. analyzed the data. C.R., B.F., E.U.W., and E.J.J. prepared the manuscript. All authors were involved in editing the manuscript.

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Data are available at <https://identifiers.org/neurovault.image:305023>